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BIOGRAPHY

Daniel Racoceanu is Professor in Biomedical Imaging, Pattern Recognition, Machine and Deep Learning, and Information/Data Analytics at Sorbonne University, and Principal Investigator at the Paris Brain Institute, within the Center for Artificial Intelligence and Data Science. His research focuses on computational pathology, spatial transcriptomics, biomedical image analysis, multimodal representation learning, and explainable artificial intelligence, with a particular emphasis on clinically meaningful and trustworthy AI systems.

He received his Ph.D. in 1997 and his Habilitation (HDR) in 2006 from the University Marie and Louis Pasteur, Besançon, France. Before joining academia, he worked as Project Manager at General Electric. In 1999, he joined the University Marie and Louis Pasteur as Associate Professor and Research Fellow at the FEMTO-ST Institute (CNRS UMR 6174).

Between 2005 and 2014, he contributed to the establishment and development of the International Joint Research Unit IPAL (UMI CNRS 2955) in Singapore, serving as its Director from 2008 to 2014. This international research structure was created through a collaboration between Sorbonne University, the French National Centre for Scientific Research (CNRS), the National University of Singapore (NUS), and the Agency for Science, Technology and Research (A*STAR). During this period, from 2009 to 2015, he was also Professor at the School of Computing at the National University of Singapore.

From 2014 to 2016, he served on the Executive Board of the University Institute of Health Engineering at Sorbonne University and was co-Director of the Cancer Theranostics research team at the Bioimaging Lab (LIB – CNRS UMR 7371, Inserm U1146).

From 2016 to 2018, he was Professor in Biomedical Imaging and Data Analytics at the Pontifical Catholic University of Peru.

Since 2020, he has been a member of the Advisory Board of the European Society for Digital and Integrative Pathology (ESDIP). He previously served as Vice-President from 2016 to 2018 and President from 2018 to 2020.

From 2018 to 2022, he served on the Board of Directors of the Medical Image Computing and Computer-Assisted Intervention Society (MICCAI). He was General Chair of MICCAI 2020 and contributed to the organization of MICCAI 2021 and MICCAI 2022.

His current work explores the integration of digital histopathology, spatial omics, and foundation AI models for next-generation precision medicine.

PROFESSIONAL & RESEARCH EXPERIENCE

Since May 2019 – **Sorbonne University, Paris Brain Institute (ICM)**, Paris, France

- **Principal Investigator, Paris Brain Institute (Institut du Cerveau – ICM)** - Inserm, CNRS, AP-HP
- **Center for Artificial Intelligence and Data Science**, Aramis Lab
- **Full Professor** @ Faculty of Science & Engineering, Engineering & Computer Science Department
    - Courses & Labs:
        - Artificial Intelligence, Machine / Deep Learning (master – Eng.) – coordinator
        - Medical and Biomedical Image Analysis (master level – Eng.) – coordinator
        - Biomedical Image Analysis (master level – Eng.) – coordinator
        - Computer Graphics (master level – Computer Sciences) – coordinator
        - Object Oriented Programming (Master level – Eng.)
        - Programming (undergraduate level – all options)
- Since 2020 – **Advisory Board** member - **ESDIP**[1] (European Society of Digital Integrative Pathology)
- 2023-2027 – Member of the **Inria Evaluation Committee** - **CE Inria** - **National Institute for Research in Digital Science and Technology**.
- 2020-2026 – PI of the Inria team-project Aramis.
- 2020 – general Chair **MICCAI** 2020[2] (Medical Image Computing and Computer Assisted Intervention)
- 2018-2022 – Member of the **Board of Directors** of the international **MICCAI Society**
- 2018-2020 – **President of ESDIP** – the European Society of Digital Integrative Pathology

2018 – 2019: Kitview, Orqual group, Lyon, France

- **Scientific Director**:
- Building and installing the R&D team @ Insavalor Lyon (INSA Lyon's industrial ecosystem).
- Setting up a CIFRE PhD in collaboration with LIRMM CNRS lab, Montpellier, France.
- Initiating an R&D collaboration (Spin-Off) in Cluj-Napoca, Romania.

2016 – 2018: **Pontifical Catholic University of Peru** (PUCP), San Miguel, Lima, Peru

- **Full Professor** @ Faculty of Science and Engineering, Department of Engineering;
- Master and PhD students' supervision @ the Medical Imaging Lab;
- European & international initiatives around Computational and Integrative Pathology;
- Succeeding to the **MICCAI 2020** bid - Medical Image Computing & Computer Assisted Intervention (MICCAI 2020 conference was hosted for the first time by Latin America);
- 2018 – Member of **MICCAI Society Board of Directors** (Medical Image Computing and Computer Assisted Intervention)

2011- 2016: **University Pierre and Marie Curie**, Paris, France

- **Full Professor** @ Engineering Department.
- 2014-2016 - **Research Director** - cancer diagnosis and therapies team, **Biomedical Image Lab** (LIB[3] - CNRS UMR 7371, Inserm U1146);
- 2008-2014 – **Director** of the **CNRS** research lab **IPAL** (UMI CNRS 2955) **Singapore**
- **Executive Board**, **University Institute of Health Engineering (IUIS) of Sorbonne University**
- Co-Director of a new Bachelor of Sciences minor: "Innovation in Public Health"
- 2016 – co-founder of the **ESDIP** (European Society of Digital Integrative Pathology)
- 2016-2018 – Vice-president of the **ESDIP** (Eur. Society of Digital Integrative Pathology).

2013 – 2015**: National University of Singapore** (**NUS**), Singapore

- **Full Professor** (adjunct) @ School of Computing, Computer Science Dpt.
- 2008-2014 – **Director** of the **CNRS** research lab **IPAL - UMI CNRS 2955 Singapore**
- NUS SoC PhD co-supervision; NUS SoC PhD reviews and jury member.

2011 – 2014**: French National Center for Scientific Research (CNRS)**

- **CNRS Research Director in Biomedical Engineering and Imaging**
- Second mandate as **IPAL UMI CNRS 2955 Director**.

2008 – 2014: **Director**, **International Joint Research Lab IPAL UMI CNRS**, Singapore

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[1] *ESDIP (European Society of Digital Integrative Pathology) : https://www.esdipath.org/*
[2] *MICCAI 2020: https://www.miccai2020.org/en/ORGANIZING-COMMITTEE.html*
[3] *Lab. Imagerie Biomédicale, LIB - UMR CNRS UMR7371, INSERM U1146: https://www.lib.upmc.fr/about-lib/?lang=en*

- Restructuration around two impactful areas: biomedical image understanding and ambient assisted living; successful CNRS et al. evaluations in 2009 and 2014; growth of the lab by bringing in the University Pierre and Marie Curie and Mines-Télécom Institute.

2009 – 2013: **National University of Singapore (NUS)**, Singapore
- **Associate Professor** (adjunct) @ **School of Computing**, Computer Science Dpt.
- NUS SoC PhD students' co-supervision, PhD reviews and PhD jurys
- Co-PI of the A*STAR SERC project "MMedWeb"
- Collaborator in a 2011 Singapore - MIT Alliance (SMA3)

2005 – 2011: **French National Center for Scientific Research (CNRS)**
- **CNRS Research Fellow (Chargé de Recherche CNRS),** creation and management of the Medical Image Indexing team at the International Research Unit PAL (UMI CNRS).
- Research focusing on Medical Image Analysis, Content Based Image Retrieval, Medical Ontologies, Medical Multimedia Fusion, Cognitive Vision.
- Participation @ CLEF international benchmark, Medical Image Indexing-Retrieval track.

1999 – 2008: **Marie and Louis Pasteur University** (former Franche-Comté Univ), Besançon, France
- **Associate Professor,** Faculty of Sciences & Technologies, Sciences & Technologies Dpt.
- Courses at Sciences and Techniques Dpt. in artificial intelligence, computer vision, diagnosis, prognosis, control sciences, fault detection, real-time, non-linear, and discrete event systems.
- Involved in the creation of the new LMD (Licence-Master-Doctorat) program.
- Research Fellow at **FEMTO-ST Institute**[4] (UMR CNRS 6174), Besançon, France, in AI applied to dynamic monitoring, diagnosis, and prognosis, in the e-maintenance framework.

1998 – 1999: **General Electric Power Generation**[5], Belfort, France (€1,2 B; 2000 pers.)
- **Project Manager**, General Electric Gas turbine department
- Design, technical, production, logistics, financial management gas turbines projects
- Reengineering, reorganization for General Electric group integration teams.
- Optimization of the management flow.
- Detection of financial niches and consolidation actions in the financial flow.
- Design of common project management and client management protocols between Alstom (old owner of the company) and General Electric (new owner of the company)

1997 – 1998: **Gaussin Manugistique**[6]**,** Héricourt, France, (M€10 /100 persons)
- Design & manufacturing of special trailers for industry, airports, and seaports.
- **Logistics & Planning Manager** - Logistics management, supply chain management
- Reengineering and management of a distributed subsidiary structure.

1993 – 1996: **University of Franche-Comté**, Univ. Institute Technology Belfort, France
- **Lecturer** (ATER - Attaché Temporaire d'Enseignement et de Recherche)
- Dpt. Production System Management, University Institute of Technology of Belfort, University of Franche-Comté, Belfort, France
- Logistic management, supply chain management courses and laboratory
- Flexible production system labs (RFID, PLC programming, production flow management); I created a set of hands-on laboratories on a state-of-the-art flexible industrial production system.

1992: **Research Institute for Welding and Material Testing**[7] (ISIM), Timisoara, Romania
- **Research Fellow** in welding robotics.

Professional Organizations

Medical Image Computing and Computer Assisted Intervention (MICCAI Society)
- 2018-2022: member of the Board of Directors and General Chair of MICCAI 20204

European Society of Digital and Integrative Pathology (ESDIP3)
- Since 2020 - ESDIP Advisory Board
- 2018 to 2020 - President & Member of the Executive Committee

[4] *FEMTO-ST Institute (UMR CNRS 6174), Besançon, France: https://www.femto-st.fr/en*
[5] *General Electric Power Generation: https://www.gevernova.com/regions/europe*
[6] *Gaussin Manugistique: http://www.gaussin.com*
[7] *National R&D Institute for Welding and Material Testing (ISIM), Timisoara, Romania: https://www.isim.ro/en/*

- 2016 to 2018 - Vice-President & Member of the Executive Committee

French Chamber of Commerce from Singapore (FCCS)

- 2011 to 2014 – co-President & co-founder of the R&D Committee of the FCCS.

Mobility across Thematic Areas

Since 2016	Integrative Digital Pathology & XAI - Prof. @ Sorbonne University and Prof. @ PUCP
2012-2016	Computational Pathology – Prof. @ UPMC & Prof. @ NUS & DR @ CNRS
2008-2012	BioMedical Imaging, Cognitive BioMedical Image, Stochastic Prior shape and Simplicial models – Prof. @ UPMC, Prof. @ NUS, DR & CR @ CNRS
2005-2008	Content-Based Medical Image Indexing and Retrieval reinforced by Semantic Approaches - A/Prof. @ UFC, A/Prof. @ NUS, CR @ CNRS
1999-2005	Dynamic Pattern Recognition for Diagnosis/Prognosis using dynamic recurrent neural networks - A/Prof. @ Marie and Louis Pasteur University (UMLP).
1992-1997	Stochastic Models Simplification and Control using Principal Component Analysis, PhD candidate @ UFC, Lecturer @ Institute of Technology, UMLP, Belfort, M.Sc. @ UTBM.
1987-1992	Mechatronics - M.Eng. @ “Politehnica” University of Timisoara (UPT).

Geographic Mobility

Since 2019	Paris, France - Prof. @ Sorbonne University, PI @ Paris Brain Institute (ICM)
2018-2019	Lyon, France - Scientific Director @ Kitview, Orqual Group
2016-2018	San Miguel, Lima, Peru - Prof. @ Pontifical Catholic University of Peru (PUCP)
2014-2016	Paris, France - Prof. @ University Pierre and Marie Curie (UPMC Univ Paris 6)
2005-2014	Singapore - CNRS Senior Research Fellow & UMI CNRS Lab Director
1999-2005	Besançon, France - A/Prof. @ Marie and Louis Pasteur University (U. Franche-Comté)
1998-1999	Belfort, France - Project Manager @ GE Energy Europe Technol. Center
1997-1998	Héricourt, France - Logistic & Planning Manager @ Gaussin S.A.
1993-1997	Belfort, France – PhD candidate @ University of Franche-Comté & Lecturer (ATER) @ Belfort-Montbéliard University Institute of Technology (IUT)
1992-1993	Belfort, France - Master of Sciences @ Univ. of Technology Belfort-Montbéliard (UTBM)
1987-1992	Timisoara, Romania - Master of Engineering @ Politehnica Univ. Timisoara (UPT)

Education

2006 **HDR (Accreditation to Supervise Research / Habilitation à Diriger des Recherches)**
- Control and Computer Science

University of Franche-Comté, Besançon, France
- Dynamic Monitoring using Artificial Intelligence Techniques

- *Keywords:* Dynamic Monitoring, Artificial Intelligence, Dynamic Neural Networks, Neuro-Fuzzy Systems, Fuzzy Petri Nets, Diagnosis, Prognosis.

1997 **Ph.D. Control and Computer Sciences, University of Franche-Comté, Besançon, France**
- Stochastic model reduction. Quasi-optimal management solution of an EDF (Electricity of France) Hydropower System (a system with 7 centrals).
- *Keywords:* Stochastic Modelling, Markov Chains, Control, Reduction Methods, Singular Perturbations - Principal Component Analysis.

(Highest distinction - jury's congratulations – très honorable avec félicitations du jury).

1993 **M.Sc. (Master of Science) Control Sciences**
University of Technology of Belfort-Montbéliard, France
(Major & with Distinction – Très Bien).

1992 **Dipl. Ing. (Master of Engineering) - Mechanical Manufacturing Engineering Technology**
Politehnica University of Timisoara, Romania
(Highest Distinction in Romanian academic system – 10/10).

Current Scientific Activity

General interest: Responsible Artificial Intelligence, High-content Biomedical Image analysis
Responsible Artificial Intelligence, Explainable AI, Computer science for biomedical data, information and knowledge analysis / management. Integrative biomedical image analysis.

Specific area of expertise: Explainable Artificial Intelligence / Deep Learning, Integrative approaches for biomedical data analysis, Semantic-driven high-content image exploration

Explainable Artificial Intelligence / Deep Learning, Integrating heterogeneous models for omics and high-content imaging data understanding, with initial (not exclusive) focus on digital pathology. Semantic-driven, traceable deep-learning and scalable image analysis.

Keywords: High-content biomedical image analysis, integrative digital pathology, semantic-driven high-content biomedical image analysis, explainable and responsible deep learning / artificial intelligence, stochastic modelling and prior shapes, mathematical morphology on sparse-sets, content-based image retrieval.

SCIENTIFIC PUBLICATIONS

Exhaustive publication list: https://daniraco.github.io/publications.html

Google Scholar: https://scholar.google.com/citations?user=2eBRLj0AAAAJ&hl=fr
Research Gate: https://www.researchgate.net/profile/Daniel_Racoceanu
ORCHID: http://orcid.org/0000-0002-9416-1803
Web of Science Researcher-ID: F-4576-2011
DBLP profile: https://dblp.org/pid/63/4416.html
Semantic Scholar ID: https://www.semanticscholar.org/author/Daniel-Racoceanu/1742773

SCIENTIFIC SUPERVISION

- **Co-guarantor Dr. Habil – habilitation to manage scientific research (HDR)**
  - Dr Nicolas LOMENIE, A/Prof, Univ Paris Descartes, IPAL Res. Fellow Jan.2009-Sep.2012
- **Post-docs supervised**
  - Dr Anuradha KAR (ICM, 2021- 2023) - Alzheimer's disease patients' stratification
  - Dr Janan ARSLAN (ICM, 2021-2023)- model. metabolic plasticity heterogeneity in melanoma
  - Dr Antoine VEILLARD (UPMC, 2013-2016) - collaborative digital pathology
  - Dr Ludovic ROUX (UJF, 2008-2014) - semantic cognitive approaches
  - Dr HUANG Chao-Hui (NUS and A*STAR, 2008-2014) - cell tracking and WSI analysis.
  - Dr Maria KULIKOVA (CNRS/UPMC 2010-2013) - marked point process, nuclei detection.
  - Dr Caroline LACOSTE (CNRS, 2005-2006) – marked point process angiography analysis
- **On-going and graduated PhDs:**
- *On-going PhD supervision:*
  - M. Mehdi HAMADACHE (start Oct. 2025, Sorbonne Univ., 100%) – spatial transcriptomics using physically inspired AI.
  - Ms. Swann RUYTER (start Oct. 2024, Sorbonne Univ., 100%) – virtual staining, virtual proteomics and virtual spatial transcriptomics using generative IA.
  - Ms. Esther KOZLOWSKI (start Oct. 2023, Sorbonne Université, 50%) – analysis of propagation pathways in Parkinson's disease using explainable AI.
  - Ms. Ayse GUNGOR (start Aug. 2023, Sorbonne Université, 50%) – eye and brain.
  - M. Ilias SARBOUT (start Oct. 2023, Sorbonne Université, 50%) – artificial vision.
- *Graduated PhDs supervised / first position after PhD:* http://www.theses.fr/fr/?q=racoceanu
  - Ms. Laura MARIN (Feb. 2025, PUCP, Lima, Pérou) – Digital Integrative Pathology for personalized medicine, R&D engineer - AREAS Grenoble, France.
  - M. Mehdi OUNISSI, (Oct. 2024, Sorbonne Univ., Paris) - Explainable Artificial Intelligence, post-doc researcher at the Rothschild Foundation, Paris, France.
  - M. Gabriel JIMENEZ, (Sept. 2024, Sorbonne Univ., Paris) - Interpretable Deep Learning in Computational Histopathology for Alzheimer Disease Patients' Stratification Refinement, Research engineer at Paris Brain Institute, Paris, France.
  - Ms. Oumeima LAIFA, (Sept. 2019, Sorbonne Univ., Paris) - A Joint Discriminative-Generative Approach for Tumor Angiogenesis Assessment Comp. Pathology, R.Eng.Owkin.
  - M. Lamine TRAORE, (Dec. 2017, UPMC, Paris) - Semantic Modelling of a Histopathology Image Exploration & Analysis Tool, CEO smart'GRAD – UPMC spin-off.
  - M. Bassem BEN CHEIKH (Sept. 26, 2017, UPMC, Paris) - Graph-based Mathematical Morphology for Characterizing the Spatial Organization of Histological Structures in High-Content Images: Appl. Tumor Microenvironment in Breast Cancer.
  - M. Olivier MORERE (June 2016, UPMC, Paris) - Deep Learning Compact and Invariant Image Descriptors for Instance Retrieval.

- Ms. Sreetama BASU (March 2015, NUS, Singapore) - Digital Reconstruction of Neuronal Structures from 3D Microscopy data / Research Fellow @ École Normale Supérieure (Ulm) Paris, Institute of Biology, Bioinformatics & Comput. Biology group;
- M. Antoine FAGETTE (June 2014, UPMC, Paris) - Dense Crowd Analysis / Research Engineer THALES Singapore;
- M. Stéphane RIGAUD (March 2014, UPMC, Paris) - Analysis-Synthesis Approach for Neurosphere Modelisation Under Phase-Contrast Microscopy / 3 years Research Fellow @ Institut Curie, Paris;
- M. Humayun IRSHAD (Jan. 2014, UJF, Grenoble) - Automated Mitosis Detection in Color and Multispectral High-Content Images in Histopathology: Appl. to Breast Cancer Grading in Digital Pathology / Res. Fellow (3 years contract) @ Harvard Medical School, Boston, US
- M. Antoine VEILLARD (Dec. 2012, NUS, Singapore) - Kernel Methods for the Incorporation of Prior-Knowledge into Support Vector Machines. Application to whole slide image analysis for breast cancer grading / Research Fellow (3 years) @ Univ. Pierre & Marie Curie, Paris;
- Ms. Roxana TEODORESCU (Apr. 2011, Univ. Besançon) - Parkinson's Disease Prognosis using Diffusion Tensor Imaging Features Fusion / Research Fellow (3 years) @ Mount Sinai Hospital, New York, USA.
- Ms. Adina TUTAC (Oct. 2010, Univ. Besançon) - Formal Representation and Reasoning for Microscopic Medical Image-Based Prognosis. Application to Breast Cancer Grading / R&D Manager @ Sionic SRL, Timisoara, Romania.
- M. Nicolas PALLUAT (Jan. 2006, Univ. Besançon) - Dynamic monitoring using temporal neuro-fuzzy syst. / Res. Fellow (3 years contract) @ Univ. Federal Santa Catarina, Brazil;
- Ms. Eugenia MICA (Sept. 2004, Univ. Besançon) - Discrete events systems monitoring using fuzzy Petri nets / A/Prof @ Univ. Valahia, Târgoviste, Romania.
- M. Ryad ZEMOURI (Nov. 2003, Univ. Besançon) - Monitoring using dynamic neural networks A/Prof. CNAM, Paris, France.

SUCCESSFUL FUNDRAISING

Major competitive projects:

- **Interdisciplinary approaches in oncogenic processes and therapeutic perspectives: Contributions of mathematics and informatics to oncology – MIC 2025 - Synergizing Mechanistic and AI Approaches for Modeling the Impact of Microenvironment Heterogeneity on Immunotherapy efficacy (SIMAI)**
  - Partners: Paris Brain Institute (CNRS UMR7225 – Inserm U1127), the University of Montpellier (LPHI - CNRS UMR5235), and the Montpellier Cancer Research Institute - IRCM (Inserm U1194).
  - Role: PI. Personal contribution: multimodal biomedical imaging analysis using frugal AI
  - Duration: 4 years (Oct. 2025 – Sept. 2029). Budget: 528 400 €.
- **Big Brain Theory (BBT3), Paris Brain Institute - STRATIFIAD: Refining Alzheimer Disease Patients' stratification using effective, traceable and explicable artificial intelligence approaches in computational histopathology.**
  - Leader of the project. Personal contribution: WSI analysis and XAI / DL approaches
  - Partners: INRIA team Aramis lab, Alzheimer team, and Data analysis center, Paris Brain Institute
  - Duration: 2,5 years (July 2021 - December 2023). Budget 200 000 €.
- **AVIESAN (French Alliance for Life and Health Sciences) - ITMO Cancer - Mathematical Approaches to Modelling Metabolic Plasticity and Heterogeneity in Melanoma (MALMO)**
  - Partners: Paris Brain Institute (CNRS UMR7225, Inserm U1127), University of Montpellier (LPHI CNRS UMR5235, LIRMM (CNRS UMR 5506) and Montpellier Cancer Research Institute - IRCM (Inserm U1194)
  - Role: PI. Personal contrib.: traceable DL for 3D histological reconstruction and tumor characterization
  - Duration: 3 years (Nov. 2020 – Oct. 2023). Budget: 400 000 €
- **EU EIT Health: PAPHOS - Platform for advanced prescriptive health operational system**
  - Granted by the European Institute of Innovation & Technology (EIT) - EIT Health
  - Big data in healthcare (digital pathology, sleep apnea)
  - URL - PAPHOS project: https://www.eithealth.eu/paphos
  - URL - EIT Health: http://eit.europa.eu/eit-community/eit-health
  - Consortium: ATHOS Spain SAE, BULL SAS, CEA, Univ. Pierre et Marie Curie, Univ. Joseph Fourier (UJF), Univ. Politécnica de Madrid, GMV, Karolinska Institutet (KI), AVENTYN.
  - Role: PI. Personal contribution: Digital Pathology use-case with Pitié-Salpêtrière Hosp., Paris.
  - Duration: 2 years (Jan. 2016 - Dec. 2018). Budget: 1.400.000 €.
- **Medical Research Foundation (FRM, France)**; Bioengineering for health

- **Multiparametric ultrasonic classification to evaluate tumor progression**
- Partners: Sorbonne Université, Hôpital Pitié-Salpêtrière, Univ. Illinois Urbana-Champaign
- Duration: 3 years (2014-2017). Role Collaborator. Personal contribution: WSI analysis using DL

- **NIH (Nat. Institute Health, US) - Advanced Ultrasonic Evaluation of Sentinel Lymph Nodes**
  - Consortium: Riverside Research, New York NY USA, Hawaii University, Honolulu, USA, Kuakini Medical Center, Honolulu, USA, Duration 5 years (2011-2016);
  - Role: collaborator. Personal contribution: WSI analysis for the classification of metastatic regions in lymph nodes, in correlation with ultrasound modalities.

- **IAMS: Integrated autonomous microscopy syst. for imaging anatomies complex 3D cell**
  - Granted by **A*STAR/JCO** Joint Council Office SERC-BMRC/A*STAR Singapore;
  - Involving IPAL and four A*STAR institutes (the Institute for Infocomm Research I2R/SERC, the Bioinformatic Institute BII/BMRC, the Institute of Molecular and Cell Biology IMCB/BMRC and the Institute of Medical Biology IMB/BMRC);
  - Duration: 3 years (2013 – 2016). Role: PI. Personal contribution: microscopic images analysis.

- **FlexMIm: Collaborative Digital Pathology**
  - Funded by the Consolidated Interministerial Fund, French Ministry of Industry (MINEFE);
  - Project involving Orange Heathcare, Tribvn (SME), Pertimm (SME), the University Pierre and Marie Curie, the Univ. Paris Diderot and the Assistance publique – Hôpitaux de Paris (AP-HP), the public hospital system of the city of Paris and its suburbs;
  - Duration: 3,5 years (2013-2016), Grant: 1.679.000 €, Pers. role: **Principal Investigator.**

- **MICO: COgnitive MIcroscope: Cognition-driven visual explorer in histopathology**
  - Granted by the Techn. for Health program (TecSan - Technologies pour la Santé) of the **French National Research Agency** (ANR - Agence Nationale de la Recherche);
  - Partners: LIP6 -UPMC, TRIBVN, THALES-TCF, Pitié-Salpêtrière Hospital, AGFA Healthcare;
  - Duration: 3,5 years (01/2011-07/2014). Grant: 1.160.000 €.
  - Personal role: **Project Director & Principal Investigator**

- **Intel. Vision Quantitative Microscopy Neural Stem Cells Progenitor Growth Differentiation**
  - Project funded by **A*STAR/JCO** (Joint Council Office SERC-BMRC/A*STAR) Singapore Involving three A*STAR institutes (Institute for Infocomm Research I2R/SERC, Bioinformatics Inst. BII/BMRC and the Inst. of Medical Biology IMB/BMRC);
  - Duration: 3,5 years (12/2009 – 05/2013). Grant: 741.000 S$. Personal role: PI.

- **C0604 - EURO-TELEPATH - "Telepathology Network in Europe"**
  - **COST (European Cooperation in Science and Technology)** Action.
  - Partners: Spain, France, Germany, Grece, Italy, Switzerland, Croatia, Finland, Lithuania, Nederland, Norway, Poland, Portugal, United Kingdom;
  - Duration: 4 years (2007-2011). Personal role: expert in semantic imaging

- **MMedWeb (Multimedia Medical Conceptual Web for Intelligent Information Access)**
  - **A*STAR – SERC** grant - Science & Engineering Research Council, Agency for Science, Technology and Research – Singapore.
  - Partners: CNRS, NUS, I2R/A*STAR, Nat. Univ. Hospital (NUH), National Healthcare Group, Singapore General Hospital (SGH);
  - Duration: 3 years (2007-2010). Grant: 530.840 S$. Personal role: **Principal Investigator.**

- **ONCO-MEDIA (Ontology and COntext related MEdical image Distributed Intel. Access)**
  - 6th **ICT-ASIA programme**
  - 11 International partners: Singapore (IPAL), France (CREATIS, LIRIS, LIP6, I3S), Switzerland (UNIGE), Taiwan (NTU), Philippine (ATENEO), Japan (CIGG);
  - Duration of the project: 2 years (2006-2007; 2009-2010);
  - Personal role: **Project Director & Principal Investigator**

AWARDS:

- 1st Prize – Poster contest – European Neuro-Ophtalmology Society EUNOS 2024, Rotterdam, Netherlands - Gungor, A., et al., Early Detection of Central Retinal Artery Occlusion within 4.5 Hours of Visual Loss: Deep Learning Method Applied on Fundus Photographs.

- 1st Place Prize - 2019 IEEE Eng. in Medicine and Biology Prize: Irshad, H. et al. (2014). Methods for Nuclei Detection, Segmentation, and Classification in Digital Histopathology: A Review - Current Status & Future Potential. *IEEE Reviews on Biomedical Engineering.*

- Best Paper Award finalist - Zemouri, R., Racoceanu, D., Zerhouni, N. (2001). A Petri nets graphic method of reduction using birth-death processes, *IEEE International Conference on Robotics & Automation - IEEE*

ICRA'2001, pp. 46-51, Seoul, Korea.

PATENTS:

- Virtual IHC “virtual staining” (paired images) — *Device and method for generating n virtual IHC stain images from one H&E image* — EP 24 305 221.4 (filed 9 Feb 2024), PCT/EP2025/053151 (filed 6 Feb 2025). Institutional co-owner (ICM, AP-HP, CNRS, INSERM, Sorbonne Univ., Univ. Paris Cité); Co-inventor (Racoceanu). PCT publication WO2025/168731 (18 Aug. 2025) Co-inventor : Daniel Racoceanu
- Virtual IHC “virtual staining” (unpaired images) — *Device and method for generating n virtual IHC stain images from one H&E image (unpaired)* — EP 24 305 224.8 (filed 9 Feb 2024), PCT/EP2025/053153 (filed 6 Feb 2025). Institutional co-owner; Co-inventor : Daniel Racoceanu
- 3D cell/particle tracking (microscopy) — *Method and system for tracking motion of microscopic objects within a 3D volume* — US 2014/0192178 A1 (pub. 10 Jul 2014, filed 13 Aug 2012). Holder: A*STAR (Singapore). Co-inventor: Daniel Racoceanu.
- Dynamic real-time monitoring system — INPI (FR) 0858450 (filed 10 Dec 2008). Holder: Em@systec (spin-off UFC). Co-inventor: Daniel Racoceanu.
- Software copyrights (France, CNRS/APP): *Mitosis Detector* (DL 05963-01, 2013); *PDFibAtl@s* (DL 04137-01, 2011); *HISTOGRAD* (DL 2944-01; APP IDDN … 14 May 2010). Author/co-author; rights held by CNRS/UMI.

CERTIFICATES:

- June 2014, Certif. in Advanced English (CAE), Cambridge English Language Assessment.